

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

In our program we plan to observe two classical cepheid variables in the M33 galaxy using photometry. With this we can compare the magnitude of the two cepheids with previous calculated light curves. This is scientifically interesting as there is a direct correlation between the period of a cepheid to its absolute magnitude, which with the apparent magnitude the distance of the cepheid from the Earth can be calculated. This can then also be used to determine the distance between multiple galaxies.

Telescope Requested (0.5 points): We are requesting to use the Bok 90" telescope.

Instrument (0.5 points): With the Bok 90" telescope we will be using the 90Prime instrument specifically with the G and I filters.

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points):

Grey is acceptable but dark is preferable, as M33 is very dim in our night sky with the cepheids having a magnitude of 19, and the brighter the moon is, the brighter the sky will be. With a brighter sky M33 will be harder to detect and our data will be less accurate, leading to an increased potential error.

Target List (2 points)

Object Name	Right Ascension	Declination	V magnitude	Object Type	Visibility
M33	01h 33m 50.89s	30° 39' 36.63"	5.72	Galaxy	June - February
D33 J013401.2+30311 3.7	01h 34m 01.1961715224s	30° 31' 13.402671456"	19.096	Classical Cepheid Variable	June – February
D33 J013352.4+30384 4.2	01h 33m 52.4402010666s	30° 38' 44.082599552"	19.380	Classical Cepheid Variable	June - February

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

A three-minute exposure was used in previous observations of M33 cepheids conducted by Maxwell Moe. We will want a SNR of at least 100 which we determined will require around 10-minute exposure for our observations. (Source:

<https://iopscience.iop.org/article/10.1086/318773/pdf>)

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)

These observations will be used in comparison to previously collected data (mentioned in the paper above) to determine if the cepheids are following the same period the paper determined. If it is the same period, we can easily determine the distance of the M33 galaxy, however, if it seems that the period has shifted, this could mean that our data is off or the star is experiencing a physical change. By determining the period of the cepheid, we can compare that to the correlation between period and cepheid absolute magnitude, which that with its apparent magnitude can be used in the distance modulus formula to calculate the distance from the Earth to the cepheid.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data.
(0.5 points)

If our plan does not work, then we will attempt to look in a different band or use archival data of the cepheids from two different dates to check to see the period is consistent between those two dates.